Neural Attention for Cross-Lingual Document Comprehension

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Abstract

We present a neural attention framework for cross-lingual document comprehension. The model aligns word embeddings across languages with a shared transformer encoder and a token-level attention scorer. On two multilingual benchmarks, our approach improves exact-match accuracy by 4.2 points over the previous state of the art, while reducing inference latency by 18 percent. We further analyze the learned attention weights and show that cross-lingual alignment emerges in the lower layers of the encoder.
Keywords: attention, transformer, cross-lingual, embedding, benchmark

1. Introduction

Understanding documents across languages is central to multilingual information retrieval and question answering. Existing systems translate the input text before comprehension, which introduces propagation errors and extra latency. In this paper we propose a single model that reads the source language directly and reasons over a shared embedding space. Our main contribution is a token-level attention mechanism that reweights source tokens according to their relevance to the query, independent of the surface language. We evaluate the framework on two public benchmarks and report both accuracy and latency.

Figure 1: Overview of the proposed attention-based framework, including the shared encoder and the token sco

2. Related Work

Cross-lingual embeddings. Multilingual word embeddings such as MUSE and LASER project several languages into one space. However, they are static and do not adapt to the document context. Neural machine translation approaches treat the problem as a translation task and lose token-level interpretability. The transformer architecture proposed by Vaswani et al. provides the backbone for our encoder, while recent work on attention visualization motivates the interpretability analysis in Section 4.

Table 1: Benchmark datasets used in the experiments, with language pairs and average document length.

2.1 Transformer basics

The transformer processes a sequence of tokens with self-attention layers, producing a contextual representation for every token. Each head computes queries, keys and values from the input, and the output is a weighted sum of the values. Multi-head attention allows the model to attend to different subspaces in parallel.

3. Method

Our framework contains three components: a shared encoder, a token-level attention scorer and a prediction head. The encoder maps tokens of any supported language into a common representation space, and the scorer computes a relevance distribution over the source tokens for a given query. The final prediction is produced by the head from the attended representation.

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score(q, k) = q * k / sqrt(d)
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3.1 Model architecture

The encoder is a 6-layer transformer with 8 attention heads and hidden size 512. We share all parameters across languages and train with a masked language modeling objective before fine-tuning on the downstream task. The attention scorer takes the query representation and the encoder outputs, and applies a bilinear score before softmax normalization.

Figure 2: Training loss curves on the two benchmarks, showing convergence within 20 epochs.

4. Experiments

We evaluate on two multilingual benchmarks: XQuAD-R and MLQA-R. We compare against three baselines: translate-then-read, static embeddings and a cross-lingual transformer without the token scorer. All models share the same parameter budget and training schedule.

4.1 Results

Our framework achieves an exact-match score of 61.3 on XQuAD-R and 58.9 on MLQA-R, outperforming the strongest baseline by 4.2 and 3.8 points respectively. The attention visualization shows that lower layers align tokens across languages, while upper layers focus on query-relevant content. Inference latency is 18 percent lower than the translate-then-read pipeline.

5. Conclusion

We presented a cross-lingual attention framework that reads the source language directly and learns token-level alignment without translation. The approach improves accuracy on two benchmarks and reduces latency. Future work includes extending the framework to more language families and document-level tasks such as summarization.

References

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